

An action floor on transverse phase-space area produces the Yang–Mills quantum-mechanical gap, and a gap forces an action unit

Correction (2026-10-02). A referee report and the October 1 audit (corpus audit §3) found an error in Proposition 5 and two weak points in the remarks. The changes, made in place below:

1. *Four dimensions.* The former Proposition 5 listed the classical coupling $1/g_{\text{cl}}^2$ among the constants and then dropped it from the $d = 4$ row, concluding that four-dimensional Yang–Mills carries “no constant at all” besides c and admits no action unit. With the gauge field as a connection, $1/g_{\text{cl}}^2$ has the dimension of action in $d = 4$, and the repository’s dimensionless coupling is $g^2 = \hbar g_{\text{cl}}^2$ (LLM.md §2). The claims “no constant at all” and “no action unit in $d = 4$ ” are **withdrawn**. Corrected row: four dimensions carry exactly one classical constant besides c , an action unit, and no length; the gap cannot enter any action monomial, and the action unit with c (and $\hbar$) yields no energy, length or time. The implication that fails in $d = 4$ is “action unit $\Rightarrow$ gap”.
2. *Classical versus quantum.* An overall factor $1/g_{\text{cl}}^2$ drops out of the classical field equations in every dimension. The former closing paragraph of Section 3 read the contrast between $d = 2, 3$ and $d = 4$ as one between classical constants. The contrast becomes physical only in the quantum weight $e^{-S_E/\hbar}$, whose coefficient $1/(\hbar g_{\text{cl}}^2)$ has dimension length $^{4-d}$; the lead, Section 3 and Section 5 now say so.
3. *Counting (Remark 3).* The former count kept one valley and an arbitrary inner cutoff x_0 ; its coefficient $16\sqrt{2}\pi m/(3g)$ of $E^{3/2} \ln E$ was two thirds of the correct one. The remark now computes the floor-admitted volume of both valleys up to a bounded remainder, $(8\sqrt{2}\pi m/g)E^{3/2}[\ln(E/\varepsilon) + O(1)]$, and compares its quotient by $\hbar^2$ with Simon’s exact leading asymptotics, which it matches, constant included.
4. *Levels (Remark 4).* The Bohr–Sommerfeld estimate describes one channel. Counting both valleys doubles every level, the coupling near the origin is ignored, and the former claim of a positive first spacing is restricted to distinct levels.
5. *Citation.* The Simon page “p. 3” was a reprint page. The passages are on pp. 210 and 212 of Ann. Phys. 146 (1983) 209–220; the constant comes from J. Funct. Anal. 53 (1983) 84–98, pp. 85–86 (both passage read on 2026-10-02). Remark 1’s “Lemma 0” of G07 now points to the oscillator bound in the proof of G07, Theorem 5.

In the Yang–Mills quantum-mechanical model of G07 the two directions of the analogy between a positive action floor $\hbar > 0$ and a mass gap can both be stated and proved inside one solved system. Forward: classical mechanics plus the single postulate that no transverse oscillation has phase-space area below $\hbar/2$ confines the escaping valley motion to $|x| \leq 2\sqrt{m} E/(\hbar g)$ at energy E and makes the admitted phase-space volume finite, $(8\sqrt{2}\pi m/g)E^{3/2}[\ln(E/\varepsilon) + O(1)]$ with both valleys counted. Its quotient by $\hbar^2$ has the leading term, constant included, of the eigenvalue counting function whose exact asymptotics Simon proved, for every positive floor value. Bohr–Sommerfeld quantization reproduces the energy unit $\varepsilon = \hbar^{4/3} g^{2/3} m^{-2/3}$ of the exact spectrum, and the operator inequality behind G07’s Theorem 5 is this postulate made rigorous by the uncertainty inequality. The abelian model has no transverse oscillation, so the same floor constrains nothing and its spectrum stays continuous: with $\hbar > 0$ assumed, a commutative field still has no gap. Backward: in the classical model with constants m and g , any positive

finite gap Δ , whatever extra constant produces it, supplies the action unit $\Delta^{3/4}m^{1/2}g^{-1/2}$, so within the model “gap” and “action unit” are dimensionally equivalent. For pure Yang–Mills the overall coupling $1/g_{\text{cl}}^2$ drops out of the classical field equations in every dimension, so the comparison across dimensions concerns the quantum weight $e^{-S_E/\hbar}$, whose coefficient $1/(\hbar g_{\text{cl}}^2)$ has dimension length^{4-d} . The equivalence holds in $d = 2$ and $d = 3$, where that coefficient carries a length. It fails in $d = 4$, the action-critical dimension. There $1/g_{\text{cl}}^2$ is itself an action and the weight carries only the pure number $1/g^2$, with $g^2 = \hbar g_{\text{cl}}^2$. An action unit therefore exists before any gap is given and determines no energy, and a gap needs a length that only dimensional transmutation generates. The floor postulate is a phase-space area statement; the necessity theorem sought in STATE item 1 is the derivation of such a floor. This is G08, exploratory; ledger row C132 cites Propositions 4–5 (Section 5).

1. The transverse phase-space area and the floor postulate

Take the scalar model of G07, Theorem 5, with classical Hamiltonian

$$H_{\text{cl}} = \frac{p_x^2 + p_y^2}{2m} + \frac{g^2}{2}x^2y^2.$$

At fixed $x \neq 0$ the transverse motion is the oscillator $E_y = p_y^2/2m + \frac{1}{2}m\omega_x^2y^2$ with $\omega_x = g|x|/\sqrt{m}$.

Proposition 1. The transverse orbit at fixed x and transverse energy E_y is the ellipse $\{p_y^2/2m + \frac{1}{2}m\omega_x^2y^2 = E_y\}$, and its enclosed phase-space area is

$$J_y = \oint p_y dy = \frac{2\pi E_y}{\omega_x} = \frac{2\pi\sqrt{m} E_y}{g|x|}.$$

Proof. The ellipse has semi-axes $\sqrt{2mE_y}$ in p_y and $\sqrt{2E_y/(m\omega_x^2)}$ in y ; the product times π gives $2\pi E_y/\omega_x$. $\square$

J_y is the adiabatic invariant of the transverse oscillation: along the valley it is conserved to leading order when ω_x changes slowly on the oscillation’s time scale, $|\dot{x}/x| \ll \omega_x$, that is $|x| \gg x_{\text{ad}}(E) = (2E)^{1/4}g^{-1/2}$ at energy E (Landau–Lifshitz, *Mechanics* §49; the note uses only Proposition 1 and the inequality below, which need no adiabatic limit).

Floor postulate F(h). No admitted state has a transverse oscillation of phase-space area below $h/2 = \pi\hbar$: $J_y \geq h/2$ whenever $x \neq 0$, and $J_x \geq h/2$ whenever $y \neq 0$.

The value $h/2$ is the Bohr–Sommerfeld ground-state area $(n + \frac{1}{2})h$ at $n = 0$; any positive multiple of h gives the same conclusions with a different pure number.

2. Forward: the floor confines the valley and produces the gap

Theorem 2. Under F(h), every admitted classical state of energy E satisfies

$$|x| \leq X(E) = \frac{2\sqrt{m} E}{\hbar g}, \quad |y| \leq X(E),$$

and the transverse energy obeys $E_y \geq \hbar\omega_x/2 = \hbar g|x|/(2\sqrt{m})$. In particular the admitted region of the energy shell is bounded, whereas without the floor the shell contains the unbounded valley motions $y = p_y = 0$, $x(t) = x_0 + p_x t/m$ of every energy $p_x^2/2m > 0$.

Proof. For $x \neq 0$, Proposition 1 and $E_y \leq E$ give $J_y = 2\pi\sqrt{m} E_y/(g|x|) \leq 2\pi\sqrt{m} E/(g|x|)$; the floor $J_y \geq \pi\hbar$ then forces $|x| \leq 2\sqrt{m}E/(\hbar g)$. Reading the same inequality as a bound on E_y gives $E_y \geq \hbar g|x|/(2\sqrt{m})$. The y bound is symmetric. $\square$

Remarks.

1. *The floor is the quantum bound.* The inequality $E_y \geq \hbar g|x|/(2\sqrt{m})$ is word for word the oscillator bound of G07, Theorem 5 (the oscillator bound in its proof): $-\frac{\hbar^2}{2m}\partial_y^2 + \frac{g^2}{2}x^2y^2 \geq$

$\hbar g|x|/(2\sqrt{m})$. The quantum theorem is the floor postulate enforced by the uncertainty inequality $\int(\mu|\phi'|^2 + \nu y^2|\phi|^2) \geq \sqrt{\mu\nu} \int |\phi|^2$, which is the statement that no normalizable state occupies a transverse phase-space area below $h/2$. So in this model the implication “ $h > 0$ gives the gap” is exact, with h entering only through the area floor, and the rest of quantum kinematics is not used.

2. *Self-consistency of the adiabatic reading.* The cutoff $X(E)$ lies in the adiabatic region: $X(E)/x_{\text{ad}}(E) = 2^{3/4}(E/\varepsilon)^{3/4}$ with $\varepsilon = \hbar^4/3 g^2/3 m^{-2/3}$, so for $E \gg \varepsilon$ the floor acts where J_y is conserved, and near the ground state, $E \sim \varepsilon$, the classical reading is only an estimate.
3. *Counting (corrected 2026-10-02).* Without the floor the phase-space volume $|\{H_{\text{cl}} \leq E\}|$ is infinite for every $E > 0$ (G07, §3.1, Mechanism; Simon, Ann. Phys. 146 (1983) 209–220, p. 210, passage read). With the floor it is finite, and it can be computed with both valleys included. Write $R = \sqrt{2E}/g$, so that the potential is below E exactly where $|xy| \leq R$; the momenta at (x, y) fill a disc of area $2\pi m(E - \frac{g^2}{2}x^2y^2)$. By Theorem 2 every admitted state lies in the box $|x|, |y| \leq X(E)$. For $X(E)^2 \geq R$, that is $E \geq \varepsilon/2$, the box carries the volume

$$P(E) = \iint_{|x|, |y| \leq X} 2\pi m \left(E - \frac{g^2}{2}x^2y^2 \right)_+ dx dy = 8\pi m ER \left[\frac{8}{9} + \frac{2}{3} \ln \frac{X^2}{R} \right] :$$

in each quadrant the strip $0 \leq x \leq R/X$, where the y -integral of $E - \frac{g^2}{2}x^2y^2$ runs to X , contributes $\frac{8}{9}ER$, and the region $R/X \leq x \leq X$, where it runs to R/x and equals $\frac{2}{3}ER/x$, contributes $\frac{2}{3}ER \ln(X^2/R)$. Inside the box the floor removes, at each (x, p_x) with $x \neq 0$, the ellipse $E_y < \hbar\omega_x/2$ of area $\pi\hbar$ in the (y, p_y) plane (Proposition 1), and at each (y, p_y) with $y \neq 0$ the ellipse $E_x < \hbar\omega_y/2$ of the same area. The admitted volume $V(E) = |\{H_{\text{cl}} \leq E\} \cap \mathbf{F}(h)|$ therefore obeys $0 \leq P - V \leq 2 \cdot \pi\hbar \cdot 2X \cdot 2\sqrt{2mE} = 16\sqrt{2}\pi m E^{3/2}/g$. With $ER = \sqrt{2}E^{3/2}/g$ and $X^2/R = 2\sqrt{2}(E/\varepsilon)^{3/2}$,

$$V(E) = \frac{8\sqrt{2}\pi m}{g} E^{3/2} \left[\ln \frac{E}{\varepsilon} + \theta(E) \right], \quad \ln 2 - \frac{10}{9} \leq \theta(E) \leq \ln 2 + \frac{8}{9}.$$

Since $\sqrt{R} = x_{\text{ad}}(E)$, the hyperbola $|xy| = R$ crosses the diagonal at the adiabatic scale of Remark 2. To leading order V is twice the one-valley integral of $(8\pi/3)\sqrt{2}mE^{3/2}/(g|x|)$, the transverse ellipse area $2\pi(E - p_x^2/2m)/\omega_x$ integrated over $|p_x| \leq \sqrt{2mE}$, over $x_{\text{ad}}(E) \leq |x| \leq X(E)$. The former version of this remark integrated one valley from a fixed inner cutoff x_0 and found the coefficient $16\sqrt{2}\pi m/(3g)$ of $E^{3/2} \ln E$, two thirds of the correct $8\sqrt{2}\pi m/g$. A floor κh with any $\kappa > 0$ in place of $h/2$ replaces X by $X/(2\kappa)$, leaves the bound on $P - V$ unchanged (the excluded ellipses grow by 2κ as the box shrinks by $1/(2\kappa)$) and shifts both bounds on θ by $-\frac{4}{3} \ln(2\kappa)$, so the coefficient of $E^{3/2} \ln E$ is independent of the floor value.

Comparison with the exact count. For the operator $-\Delta + |xy|^\alpha$, Simon proves the limit $E^{-1-1/\alpha}(\ln E)^{-1}N(E) \rightarrow \pi^{-1}$ (J. Funct. Anal. 53

- (1983) 84–98, Theorem 1.4 and the display after Theorem 1.5, pp. 85–86, passage read). At $\alpha = 2$ this is $N(E) \simeq \pi^{-1}E^{3/2} \ln E$, the growth also stated in Ann. Phys. 146, p. 212, end of Section 2. The dilation $x \mapsto \ell x$ with $\ell^6 = \hbar^2/(mg^2)$ carries $H = -\frac{\hbar^2}{2m}\Delta + \frac{g^2}{2}x^2y^2$ to $\frac{\varepsilon}{2}(-\Delta + x^2y^2)$, because both coefficients become $\hbar^2/(2m\ell^2) = g^2\ell^4/2 = \varepsilon/2$. Hence $N_H(E) \simeq (2\sqrt{2}/\pi)(E/\varepsilon)^{3/2} \ln(E/\varepsilon)$, while the floor count gives $V(E)/h^2 = (2\sqrt{2}/\pi)(E/\varepsilon)^{3/2}[\ln(E/\varepsilon) + \theta(E)]$. The leading terms agree, constant included. The identification $N \approx V/h^2$ is the Weyl heuristic, and the agreement is a comparison of two computed leading terms; Simon’s proof runs through a Feynman–Kac lower bound and sliced-bread upper bounds. Because the coefficient is independent of the

floor value, the agreement tests that a positive area floor exists and leaves the number $h/2$ untested. Simon (p. 86) calls a general geometric interpretation of the constant π^{-1} an open question; no search was run for later literature that supplies one, and no novelty is claimed.

4. *Semiclassical levels.* With $J_y = (n + \frac{1}{2})h$ conserved, the valley motion has effective Hamiltonian $p_x^2/2m + \kappa_n|x|$, $\kappa_n = (n + \frac{1}{2})\hbar g/\sqrt{m}$, and Bohr–Sommerfeld quantization $\oint p_x dx = (N + \frac{1}{2})h$ with $\oint p_x dx = \frac{8\sqrt{2m}}{3}E^{3/2}/\kappa_n$ gives

$$E_{n,N}^{\text{BS}} = \varepsilon \left[\frac{3\pi(2n+1)(2N+1)}{16\sqrt{2}} \right]^{2/3}, \quad \varepsilon = \hbar^{4/3} g^{2/3} m^{-2/3}.$$

The unit ε is the exact one of G07, Theorem 5(4), as Theorem 1 of G07 requires; the pure numbers are semiclassical estimates of e_n and are labelled as such. The estimate describes one channel, the x -axis with its two half-valleys. The y -axis gives the same levels, so counting both channels doubles every level, and the coupling of the channels and of different n near the origin, where the adiabatic reading fails (Remark 2), is ignored (corrected 2026-10-02). The estimate therefore shows a discrete spectrum with a positive lowest level and positive distances between distinct levels: in that sense the old quantum theory of this model already has a gap. The splitting of the doublets and the simplicity of the ground state, which the Jaffe–Witten conditions need, come from G07, Theorem 5(3). As a consistency check, each channel and each n contribute $\frac{4\sqrt{2}}{3\pi}(E/\varepsilon)^{3/2}/(n+\frac{1}{2})$ levels below E ; summed over both channels and over n up to the adiabatic limit $n_* \simeq 2^{-1/4}(E/\varepsilon)^{3/4}$, where the turning point E/κ_n reaches $x_{\text{ad}}(E)$, they give $(2\sqrt{2}/\pi)(E/\varepsilon)^{3/2} \ln(E/\varepsilon)$ to leading order, the same leading term as Remark 3.

Corollary 3 (the commutative model). For the abelian matrix model, and for $D = 1$, the potential vanishes identically: there is no transverse oscillation, $F(h)$ imposes no condition, the classical energy shells stay unbounded and the quantum spectrum is $[0, \infty)$ (G07, Theorem 6(5)). An action floor produces a gap only in the presence of the commutator potential, which is what supplies the transverse ellipses whose area the floor bounds.

For the $\text{SU}(2)$ model with $D \geq 2$ the same argument applies to each of the two transverse directions of $\vec{x}_j$ relative to $\vec{x}_i$: the floor gives $E_\perp \geq \hbar g |\vec{x}_i|/\sqrt{m}$, the bound of G07, Theorem 6(1), and $|\vec{x}_i| \leq \sqrt{m} E/(\hbar g)$ on the energy shell.

3. Backward: a gap forces an action unit; at the action-critical dimension the unit is classical and fixes no scale

Proposition 4. Let the classical model with fixed constants m, g (dimension vectors $d_m = (1, 0, 0)$, $d_g = (\frac{1}{2}, -1, -1)$) be enlarged by any fixed constants, and suppose the enlarged theory has a positive finite gap Δ depending on the fixed constants only. Then the theory has the action unit

$$A = \Delta^{3/4} m^{1/2} g^{-1/2},$$

and conversely an action unit A gives the energy unit $A^{4/3} g^{2/3} m^{-2/3}$.

Proof. Energy has dimension vector $(1, 2, -2)$ and action $(1, 2, -1)$. The system $a d_m + b d_g + c(1, 2, -2) = (1, 2, -1)$ has the unique solution $a = \frac{1}{2}$, $b = -\frac{1}{2}$, $c = \frac{3}{4}$: the length and time components give $-b + 2c = 2$, $-b - 2c = -1$, hence $c = \frac{3}{4}$, $b = -\frac{1}{2}$, and the mass component gives $a = \frac{1}{2}$. So $m^{1/2} g^{-1/2} \Delta^{3/4}$ has action dimension whatever the extra constants are. The converse is G07, Theorem 3 with $k = 4$. $\square$

Within the model, therefore, “the gap is positive” and “an action unit exists” are the same dimensional statement, and G07’s Theorem 5 supplies the dynamical content, $0 < \delta_1 < \infty$, that turns the unit into the number $\Delta = \delta_1 \varepsilon$.

Proposition 5 (which field theories share this equivalence; corrected 2026-10-02). Write the Euclidean Yang–Mills action with the gauge field as a connection, $D = \partial + A$, $[A] = \text{length}^{-1}$, $x_0 = ct$:

$$S_E = \frac{1}{4g_{\text{cl}}^2} \int F_{\mu\nu}^a F_{\mu\nu}^a d^d x, \quad [1/g_{\text{cl}}^2] = \text{action} \cdot \text{length}^{4-d}.$$

The weight $e^{-S_E/\hbar}$ depends on the coupling through $\lambda_d = \hbar g_{\text{cl}}^2$, of dimension length^{d-4} ; in $d = 4$ the repository writes $g^2 = \hbar g_{\text{cl}}^2$, a pure number (LLM.md §2; G07, Proposition 10, uses the trace normalization of the same action). With the speed c and a gap Δ :

1. In $d = 2$ and $d = 3$ the pair $(1/g_{\text{cl}}^2, c)$ contains neither an action unit nor an energy unit. Adjoining Δ gives exactly one action unit, adjoining $\hbar$ gives exactly one energy unit, and the equivalence of Proposition 4 holds.
2. In $d = 4$, $1/g_{\text{cl}}^2$ is itself an action unit. The constants $(1/g_{\text{cl}}^2, c, \Delta)$ are dimensionally independent and their only action monomial is $1/g_{\text{cl}}^2$, with Δ at exponent zero. The constants $(1/g_{\text{cl}}^2, c, \hbar)$ contain no energy, length or time unit. So “gap $\Rightarrow$ action unit” holds trivially and “action unit $\Rightarrow$ gap” fails.

d	classical constants (with c for Yang–Mills)	action unit with Δ adjoined	energy unit with $\hbar$ adjoined
2	$1/g_{\text{cl}}^2 \sim (1, 4, -1)$	$(1/g_{\text{cl}}^2)^{1/3} c^{-2/3} \Delta^{2/3}$	$\hbar^{3/2} g_{\text{cl}} c = \hbar c \lambda_2^{1/2}$
3	$1/g_{\text{cl}}^2 \sim (1, 3, -1)$	$(1/g_{\text{cl}}^2)^{1/2} c^{-1/2} \Delta^{1/2}$	$\hbar^2 g_{\text{cl}}^2 c = \hbar c \lambda_3$
4	$1/g_{\text{cl}}^2 \sim (1, 2, -1)$, an action	$1/g_{\text{cl}}^2$ itself; Δ enters with exponent 0	none: $\hbar g_{\text{cl}}^2 = g^2$ is a pure number
mechanics, $k = -2$	$m, \lambda \sim (1, 4, -2)$	$\sqrt{m\lambda}$ already classical; Δ adds nothing	none: $2m\lambda/\hbar^2$ is a pure number

Proof. Dimension vectors are over (mass, length, time); action is $(1, 2, -1)$, energy $(1, 2, -2)$, $c \sim (0, 1, -1)$. Solve $a d_{1/g_{\text{cl}}^2} + b d_c + c' (1, 2, -2) = (1, 2, -1)$. For $d = 3$, adding the length and time equations gives $2a = 1$, then $c' = \frac{1}{2}$, $b = -\frac{1}{2}$; for $d = 2$ the sum gives $3a = 1$, then $c' = \frac{2}{3}$, $b = -\frac{2}{3}$; for $d = 4$ it gives $a = 1$, then the mass equation gives $c' = 0$ and the length equation $b = 0$. The determinants of the three dimension vectors are $-3, -2, -1$ for $d = 2, 3, 4$, so each solution is unique. With $\hbar \sim (1, 2, -1)$ in place of Δ the determinants are -2 and -1 for $d = 2, 3$, and the displayed energy monomials have vector $(1, 2, -2)$ by direct addition. In $d = 2, 3$ the pair $(1/g_{\text{cl}}^2, c)$ alone fails: the mass and length equations fix $a = 1$ and $b = -2$ (respectively $b = -1$), and the time component is then 1 (respectively 0), never -1 or -2 . In $d = 4$, $1/g_{\text{cl}}^2$ and $\hbar$ have the same vector, so the span is that of $(1, 2, -1)$ and $(0, 1, -1)$; $a(1, 2, -1) + b(0, 1, -1) = (1, 2, -2)$ forces $a = 1$ (mass), $b = 0$ (length) and then $-1 = -2$ (time), and the targets $(0, 1, 0)$ and $(0, 0, 1)$ fail at the time component in the same way. The $k = -2$ row is G07, Section 6, Proposition 13. $\square$

In $d = 2, 3$ the quantum theory in infinite volume has the constants $(\hbar, c, \lambda_d)$, which are independent, so a gap that depends on them alone is $\Delta = \kappa \hbar c \lambda_d^{1/(4-d)}$ with a pure number κ (G07, Theorem 1). The action unit of the table is then $\kappa^{2/3} \hbar$ in $d = 2$ and $\kappa^{1/2} \hbar$ in $d = 3$: the gap returns $\hbar$ up to a pure number.

Two facts fix the reading of the table. First, the overall factor $1/g_{\text{cl}}^2$ drops out of the classical field equations $D_\mu F_{\mu\nu} = 0$ in every dimension (G07, Proposition 10), and the dilation $A \mapsto sA(sx)$ maps solutions to solutions in every d while multiplying the action by s^{4-d} . Classical

Yang–Mills therefore has no physical scale in any dimension, and $1/g_{\text{cl}}^2$ only converts the geometry of a solution into action and energy. The difference between the rows becomes physical through the quantum weight $e^{-S_E/\hbar}$, whose coefficient $1/\lambda_d = 1/(\hbar g_{\text{cl}}^2)$ has dimension length^{4-d} . In $d = 2, 3$ the weight carries a length and, with c , an energy, and a gap is a pure number times that energy. In $d = 4$ the weight carries the pure number $1/g^2$ and no length. Second, $d = 4$ is action-critical in the sense of G07, Section 6, and shares its row type with the inverse-square potential: the classical constants own an action unit and no energy unit. This is the dimensional reason the four-dimensional problem is one-directional. A gap needs a length that the constants of the weight do not contain; dimensional transmutation generates it from the regulator (G07, Proposition 10), and the gap then returns no action unit beyond the $1/g_{\text{cl}}^2 = \hbar/g^2$ already present. Along the continuum trajectory of asymptotic freedom $g^2(a) \rightarrow 0$, so this bare action unit grows without bound in units of $\hbar$; the continuum theory keeps $\hbar$, c and the transmuted scale, the dimensionless coupling having been traded for that scale. The same remark applies to Proposition 4: the trajectories of H_{cl} depend on g^2/m alone and m normalizes the action, so the equivalence there is also a statement about the quantized model.

4. What this decides for the positive-action question

- **Direction.** In the solved model the analogy is an equivalence at the level of units and an implication at the level of dynamics: the floor postulate, used only through the transverse area, produces the gap (Theorem 2 and Remark 1), and the gap returns the unit (Proposition 4). The mass-gap problem is the forward direction with $\hbar$ and the gauge group given; the necessity problem of STATE item 1 is the derivation of the floor itself. Inside this model those two problems have the same dimensional content, and the necessity problem is exactly as hard as proving a gap from a consistency premise that does not name $\hbar$.
- **The second ingredient.** Corollary 3 states the commutative case with $h > 0$ assumed: an action floor alone gives no gap; the commutator potential is what creates transverse oscillations for the floor to act on. Any transported statement “ $h > 0$ gives a gap” must name the structure that plays this role.
- **Areas.** The floor is a bound on a phase-space area, the ellipse of Proposition 1. The programme’s anchor, Galileo’s area $vF\tau^3/(6m)$ between the inertial line and the parabola, is a configuration-space area converted to action by the factor $3F/v$ (N01 §1). The model shows what a successful floor would have to look like on the phase-space side: a lower bound on the area of a closed transverse orbit, invariant under the adiabatic deformation of the orbit. A constant-force motion has no closed transverse orbit, which is one more statement of why the Galileo class is similarity-covariant (G07, Proposition 14).
- **Gate.** For gap-track work the supplied/generated gate of G07 can be applied in phase-space terms: a generated gap needs a family of closed orbits whose area can fall below h along a non-compact direction.

5. Strategic consequence

G08 closes the loop inside the Yang–Mills quantum-mechanical model: floor gives gap, gap gives unit, and the commutative model is inert under the floor. Three facts are retained for the programme. First, the equivalence of gap and action unit holds when the coefficient of the quantum weight, $1/(\hbar g_{\text{cl}}^2)$, carries a length ($d = 2, 3$, and the matrix model with its constants m , g). In $d = 4$ the classical coupling is already an action unit, the weight carries a pure number, and a gap requires a transmuted length; the contrast between the dimensions is a quantum one, since $1/g_{\text{cl}}^2$ drops out of the classical field equations in every dimension. (Corrected 2026-10-02: the earlier text said $d = 4$ carries no classical constant and no action unit.) Second, the phase-space area of a closed transverse orbit is the object a floor must bound. Third, the floor-admitted classical volume divided by h^2 reproduces the leading eigenvalue asymptotics of

the model, constant included, for every positive floor value (Remark 3). STATE item 1's target stays the derivation of the floor from consistency premises; N02 gains the phase-space form of the floor as the statement to aim at. Ledger row C132 cites Propositions 4–5; its clause “none in $d = 4$ ” needs the correction of Proposition 5: in $d = 4$ the classical coupling is itself an action unit, the gap cannot enter it, and it yields no energy unit. Theorem 2, Propositions 1, 4, 5 and the volume formula of Remark 3 are elementary. The Weyl comparison of Remark 3 and the Bohr–Sommerfeld levels of Remark 4 are labelled estimates. Sources are in B79; the J. Funct. Anal. passage of Remark 3 was read on 2026-10-02 and is cited inline.